Customer Segmentation Analysis

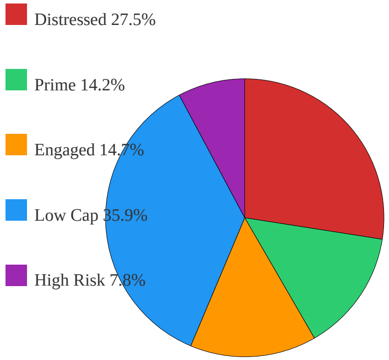
Data-Driven Portfolio Strategy | 5 Customer Segments | Confidential

9,417	**18**	**k = 5**	**2.5-3.5x**
Customers	Raw Features	Optimal Clusters	Expected ROI

Executive Summary

5 Customer Segments | Portfolio Risk & Revenue Overview

Portfolio Distribution



63.4% of portfolio in at-risk segments (Distressed + Low Cap)

Segment Financial Impact

Segment	Size	Avg Bal	Risk	Rev/Cust
Distressed	2,458 (27.5%)	$2,725	HIGH	-$75-$125
Prime	1,269 (14.2%)	$178	LOW	+$400-$500
Engaged	1,312 (14.7%)	$3,117	MOD	+$250-$350
Low Cap	3,210 (35.9%)	$698	LOW	$20-$50
High Risk	701 (7.8%)	$1,068	MED-HI	$75-$125

Strategic Priorities

Priority	Initiative	Investment	ROI
1 URGENT	Distressed Intervention	$305-355K	3-5x
2 URGENT	High Risk Prevention	$295K	3-5x
3 HIGH	Prime Growth	$405K	2.4x
4 HIGH	Engaged Optimization	$295K	2.5x
5 EFF	Low Cap Activation	$155K	4.7x

Data Overview

9,417 Customers | 18 Raw Features | Snapshot Data

9,417	**18**	**8,950**	**467 (4.9%)**	**313**	**99.8%**
Total Customers	Features	After Cleaning	Outliers Removed	Missing Imputed	Data Quality

Feature Breakdown

Category	Features	Count	Business Relevance
Balance & Capacity	BALANCE, CREDIT_LIMIT, BALANCE_FREQUENCY	3	Credit usage & capacity headroom
Purchase Behavior	PURCHASES, ONEOFF_PURCHASES, INSTALLMENTS, PURCH_FREQ, PURCH_TRX	5	Spending activity & patterns
Cash Advance	CASH_ADVANCE, CASH_ADV_FREQUENCY, CASH_ADV_TRX	3	Emergency borrowing signal
Payment Behavior	PAYMENTS, MINIMUM_PAYMENTS, PRC_FULL_PAYMENT	3	Creditworthiness indicator
Misc / Tenure	ONEOFF_FREQ, INSTALL_FREQ, TENURE	3	Behaviour & relationship age

Key Stats: Avg Credit Limit $4,571 | Avg Balance $1,564 | Median Balance $627 | Avg Tenure 12 months | Full-Payment Rate 23% | No date columns - snapshot / cross-sectional data

Data Quality & Preparation

Cleaning | Imputation | Feature Engineering

Slide 4

Step	Action	Before	After	Quality Impact
1 Outliers	Isolation Forest (contamination=0.05)	9,417 customers	8,950 customers	467 anomalies removed (4.9%)
2 Missing	Mean imputation (MINIMUM_PAYMENTS)	314 missing (3.3%)	0 missing	Full dataset coverage
3 Scaling	StandardScaler z-score normalisation	Raw $ amounts	Standardized features	Equal feature weighting
4 Engineering	7 derived business metrics	18 raw features	7 engineered metrics	Business-aligned clustering

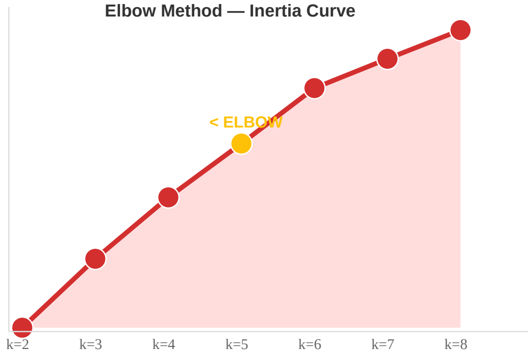
7 Engineered Business Metrics

Metric	Formula	High = ...	Low = ...
Utilization Rate	Balance / Credit Limit	Financial stress (>70%)	Healthy capacity (<20%)
Payment Quality	% Full Payments	Creditworthy borrower	Default risk signal
Engagement Score	Avg(Purchase Freq, Cash Adv Freq)	Active, loyal user	Churn risk
Payment Consistency	Actual Payments / Minimum	Aggressive paydown	Min-only payer
Debt-to-Limit	Balance / Credit Limit	>70% stress signal	<20% healthy
Tenure Score	Months / 12	Long-term relationship	New account
Transaction Volume	Purchase TRX + Cash Adv TRX	High activity	Inactive account

Methodology

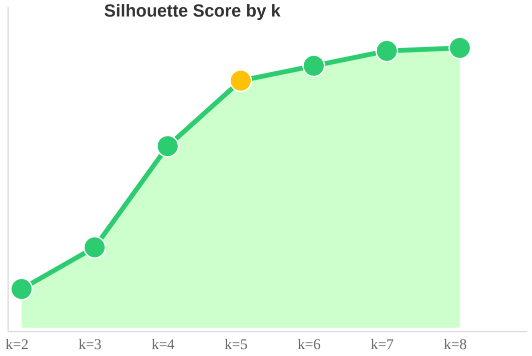
K-Means | Elbow Method + Silhouette Score => Optimal k = 5

Elbow Method (Inertia)



k	Inertia	% Impr	
2	49,005	-	
3	41,942	14.4%	
4	35,645	15.0%	
5	30,127	15.5%	< ELBOW
6	24,443	18.9%	Post-elbow
7	21,439	12.3%	Diminishing

Silhouette Score



k	Score	Quality
2	0.283	Poor
3	0.297	Poor
4	0.331	Fair
5	0.353	Good (Chosen)
6	0.358	Good
8	0.364	Peak +3%

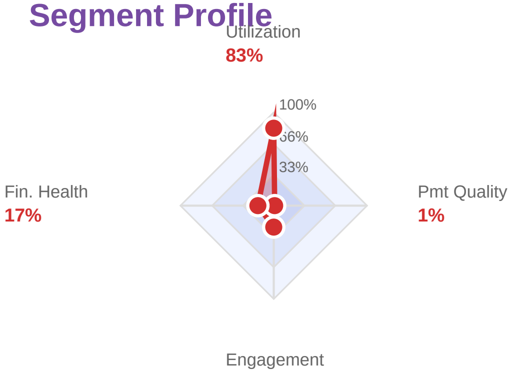
k=8 peaks at 0.364 - only +3% better than k=5 but requires 3 extra management strategies

Consensus: k=5 is Optimal. Elbow Method shows clear inflection at k=5 (15.5% improvement => 12.3% drop-off at k=7). Silhouette k=5 = 0.353 vs k=8 = 0.364 - marginal +3% does not justify 60% added operational complexity.

Distressed Revolvers

2,458 Customers (27.5%) | Avg Balance $2,725 | Risk: HIGH | Action: Intervene

Segment Profile



KPI Snapshot

Avg Balance	**$2,725**
Utilization	**83%**
Full Pmt Rate	**0.9%**
Engagement	**0.23**
Default Risk	**15-20%**
Portfolio Exposure	**$6.7M**

Key Metrics

Metric	Value	Business Implication
Utilization Rate	83%	Near credit limit — severe financial stress
Full Payment Rate	0.9%	Minimum-only payments — default accumulation risk
Engagement Score	0.23	Avoidance behaviour, not disinterest
Aggregate Balance	$6.7M	Critical single-segment portfolio exposure
Est. Default Rate	15-20%	70-90% above portfolio baseline
Pmt Consistency	1.57x min	Barely above minimum threshold

Business Strategy

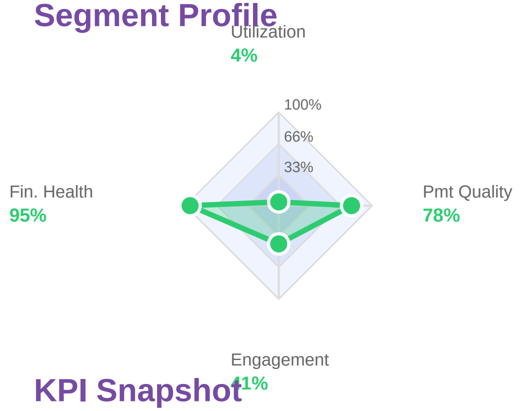
Phase 1 (Wk 1-2)	Risk triage: Recoverable (40%), Declining (40%), Terminal (20%). Build automated scoring dashboard. Do not treat the segment as homogeneous.
Phase 2 (Wk 3-8)	Hardship programmes, balance transfers at lower rates, credit counselling, income verification.
Phase 3 (Wk 9+)	Freeze limit increases; auto-payment enrolment; SMS/email alerts at 3 and 7 days pre-due.

Investment: **$305-355K/yr** Expected Benefit: **$1-2M loss prevention** ROI: **3-5x**

Prime Customers

1,269 Customers (14.2%) | Avg Balance $178 | Risk: LOW | Action: Grow & Retain

Segment Profile



KPI Snapshot

Avg Balance	**$178 (4.4%)**
Full Pmt Rate	**77.6%**
Engagement	**0.41**
Pmt Consistency	**16.4x min**
Annual Revenue	**$400-500**
Lifetime Value	**$3,000-5,000**

Key Metrics

Metric	Value	Opportunity
Utilization	4.4%	Massive headroom for earned access expansion
Full Payment	77.6%	Highest creditworthiness signal in portfolio
Monthly Spending	$146	Strategic, disciplined card usage
LTV Estimate	$3-5K	Premium long-term value — churn is structural loss
Pmt Consistency	16.4x min	Financially disciplined; no repayment risk
Rev/Customer	$400-500	Highest profitability in portfolio

Business Strategy

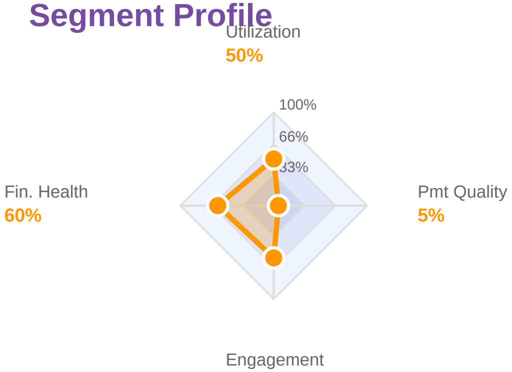
Phase 1 (Wk 1-2)	VIP recognition and dedicated service access before any cross-sell. Relationship anchoring reduces price sensitivity.
Phase 2 (Wk 3-8)	Premium card tier (2-3% cashback); personal loans at 2-3% below market; credit-limit increases tested with holdouts.
Phase 3 (Wk 9+)	Annual events; $100 referral incentives; quarterly business reviews; personalised roadmap.

Investment: **$405K/yr** Expected Benefit: **$600K-$1M revenue growth** ROI: **3-4x first year, 6x+ sustained**

Engaged Optimizers

1,312 Customers (14.7%) | Avg Balance $3,117 | Risk: MODERATE | Action: Optimize

Segment Profile



KPI Snapshot

Monthly Spending	**$242 (HIGHEST)**
Engagement	**0.56 (PEAK)**
Utilization	**49.9%**
Full Pmt Rate	**4.6%**
Pmt Consistency	**5.12x min**
Annual Revenue	**$250-350**

Key Metrics

Metric	Value	Insight
Monthly Spending	$242	Most active cohort — highest purchase frequency
Engagement Score	0.56	Peak engagement; stickiest segment after Prime
Balance	$3,117 (49.9%)	Deliberate balance management, not stress
Pmt Consistency	5.12x min	Methodical repayment — not a default risk
Churn Risk	LOW	High engagement is the strongest retention anchor
Rev/Customer	$250-350	Reliable, recurring revenue base

Business Strategy

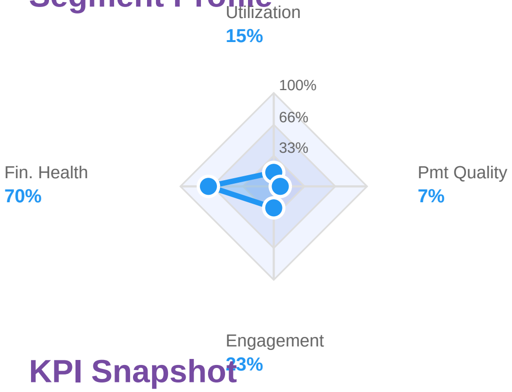
Phase 1 (Wk 1-4)	Enhanced rewards: 2x points on first $500/month; partner-merchant bonuses. Frame as a spending tool, not a debt solution.
Phase 2 (Wk 5-8)	0% APR installment plans; balance-transfer campaigns; debt-consolidation offers. Do not apply Distressed intervention playbook here.
Phase 3 (Wk 9+)	Business Owner Circle: community, monthly webinars, personalised spending insights.

Investment: **$295K/yr** | Expected Benefit: **+$1.3M balance growth + improved retention** | ROI: **4-5x**

Low Capacity / Disengaged

3,210 Customers (35.9%) | Avg Balance $698 | Risk: LOW | Action: Activate or Exit

Segment Profile



KPI Snapshot

Monthly Spending	**$48 (LOWEST)**
Engagement	**0.23 (LOWEST)**
Utilization	**15.1%**
Annual Revenue	**$20-50**
Pmt Consistency	**6.9x min**
Lifetime Value	**$300-600**

Key Metrics

Metric	Value	Challenge
Monthly Spending	$48	Portfolio floor; mostly inactive accounts
Engagement Score	0.23	Lowest engagement — high churn vulnerability
Utilization	15.1%	Minimal credit usage; no active relationship
Annual Revenue	$20-50	May not cover cost-to-serve ($60-80 est.)
Churn Risk	HIGH	Low stickiness — easy to lose without notice
LTV	$300-600	Portfolio drag without activation

Business Strategy

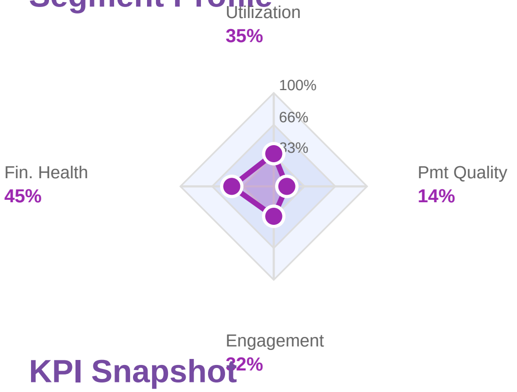
Activate 50%	Win-back: 5% cashback on groceries/gas for 3 months; micro-incentives. Target clean dormant accounts only — suppress payment-stressed customers.
Digitise 30%	Auto-paydown, digital servicing, automated statements to reduce cost-to-serve. Calculate account P&L; before any offer.
Exit 20%	After 90-day activation attempt: close/downgrade non-responsive accounts. Estimated $500K+ cost savings.

Investment: **$155K/yr** Expected Benefit: **$200K revenue + $500K cost savings** ROI: **4.7x (cost reduction focus)**

High Risk / New Customers

701 Customers (7.8%) | Avg Balance $1,068 | Risk: MEDIUM-HIGH | Action: Prevent Deterioration

Segment Profile



KPI Snapshot

Tenure	**0.6 yrs (NEW)**
Balance	**$1,068 (34.9%)**
Full Pmt Rate	**14.1%**
Engagement	**0.32**
Deterioration Risk	**50-60%**
Prevention ROI	**10:1**

Key Metrics

Metric	Value	Risk Signal
Tenure	0.6 yrs	Critical early-relationship window active now
Balance	$1,068 (34.9%)	Emerging stress — early deterioration
Full Payment	14.1%	Payment difficulty emerging; not yet acute
Deterioration Risk	50-60%	Will migrate to Distressed without action
Prevention ROI	10:1	Intervening now is 10x cheaper than curing later
Default Prob.	+15-25% above base	Elevated and rising without intervention

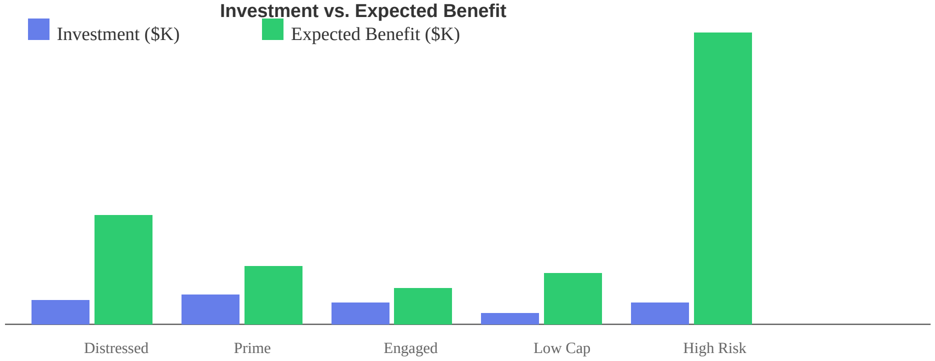
Business Strategy

Phase 1 (Wk 1-2)	Early-warning scoring: flag late payments, usage spikes, balance acceleration. Daily automated alerts. Do not wait for 30+ DPD.
Phase 2 (Wk 3-8)	Proactive financial check-ins; hardship conversations; income verification; assistance referrals. Frame as service, not surveillance.
Phase 3 (Wk 9+)	Credit builder: auto limit-increases for consecutive on-time payments; rate reductions tied to repayment behaviour; financial literacy tools.

Investment: **$295K/yr** Expected Benefit: **$3-5M default prevention** ROI: **3-5x**

Recommendations & 90-Day Plan

Implementation Roadmap | Total Investment $1.45-1.50M => Expected Benefit $2.2-4.7M



Initiative	Priority	Investment	Expected Benefit	ROI	Owner
Distressed Intervention	URGENT — Month 1	$305-355K	$1-2M loss prevention	3-5x	Risk Team
High Risk Prevention	URGENT — Month 1	$295K	$3-5M default prevention	3-5x	Risk Team
Prime Growth	HIGH — Month 2	$405K	$600K-$1M revenue	2.4x	Product
Engaged Optimization	HIGH — Month 2	$295K	$400-600K revenue	2.5x	Marketing
Low Cap Activation	EFF — Month 3	$155K	$200K rev + $500K savings	4.7x	Operations
TOTAL PORTFOLIO	**Months 1-3**	**$1.45-1.50M**	**$2.2-4.7M benefit**	**2.5-3.5x**	**Leadership**

Monthly KPIs: Default rate by segment | Early-warning detection rate (target 70%) | Prime upgrade rate (target 40%) | Engaged balance growth (+15% YoY) | Churn: Prime >95%, Engaged >92% | NPS by segment